

the basic Git workflow. As part of this workflow, we will perform the following steps in the order they are listed:

1. Download source code from the remote repository
2. Inspect the repository
3. Modify the repository and check the status of the repository
4. Upload files to the repository
5. Sync local repository to the remote repository

Download source code

Clone is a Git operation that downloads the remote repository to the local file system. For our demonstration, we will use the *Redis* repository hosted on GitHub. This repository is publicly accessible. To clone this repository, execute the command given below in the terminal:

```
$ git clone https://github.com/redis/redis.git
```

In the above command, the *clone* is Git's sub-command, and the next argument is the unique URL of the repository. After successful completion, you should see an output similar to this:

```
Cloning into 'redis'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 69075 (delta 1), reused 2 (delta 0), pack-reused 69064
Receiving objects: 100% (69075/69075), 92.93 MiB | 1.65 MiB/s, done.
Resolving deltas: 100% (48902/48902), done.
Updating files: 100% (837/837), done.
```

Execute the *ls* command as follows to verify the download of the repository:

```
$ ls redis/
```

This command should generate the following output:

```
00-RELEASENOTES  deps/      README.md
runtest-moduleapi* tests/
BUGS             INSTALL    redis.conf
runtest-sentinel* TLS.md
CONTRIBUTING    Makefile  runtest*
sentinel.conf    utils/
COPYING          MANIFESTO runtest-
cluster*        src/
```

Inspect the repository

Now that we have a copy of the remote repository on a local file system, let us inspect the repository. First, *cd* to *redis* as follows:

```
$ cd redis
```

To list all commits from the repository, execute the command:

```
$ git log
```

This command will show details of the commit. The output will appear in the UNIX pager (e.g., *less*). One can use the up and down arrow keys to scroll through the commits. Press *q* to quit the pager.

The *git log* command generates a verbose output. Execute the command given below to generate the non-verbose output:

```
$ git log --oneline
```

This command should generate the following output:

```
57b8e0069 support sentinel myid subcommand (#7858)
bea40e6a4 memory reporting of clients argv (#7874)
f87a47dc9 fix new build warning about getGenericCommand (#7882)
6418d767e redis-cli add control on raw format line delimiter (#7841)
66ee45b65 Add GET parameter to SET command (#7852)
```

In the above command, the first column is a unique SHA-1 hash of the commit. This value is generated and assigned to each commit automatically.

In Git terminology, it is called a *commit id*. The second column is the commit message, which was provided by the user. We can use the *commit id* to check commit contents. For example, the command given below shows the contents of the '57b8e0069' commit:

```
$ git show 57b8e0069
```

This command will show details of commit like SHA-1 hash, the author's name and e-mail, date and time, commit message, and the actual code changes made. One can use the up and down arrow keys to scroll through the commit and *q* to quit the UNIX pager.

To see files belonging to the particular commit, use the *--name-only* option as follows:

```
$ git show 57b8e0069 --name-only
```

This command will generate the following output:

```
commit 57b8e00695d037971
342cbcc43734a990bfaf081
Author: Wen Hui <wen.hui.ware@gmail.com>
Date: Mon Oct 5 06:37:20 2020 -0400

    support sentinel myid subcommand
    (#7858)

src/sentinel.c
```

The above output shows that the *src/sentinel.c* file belongs to the '57b8e0069' commit id.

Modify the repository

We can modify the repository by adding, removing, or editing files from the repository. Each operation will change the status of the repository. Let us verify the status by executing the command:

```
$ git status -s
```

This command will not generate any output, as the repository status is clean.